

Count-form inference sanity check: Fleiss (1971)

Package versions: `misskappa` 1.0.0, `irrCAC` 1.0.

Direct comparison on `dat.fleiss1971`

We compute Fleiss' kappa from Fleiss-form count data and form the same Wald interval $\hat{\kappa}_F \pm z_{0.975} \widehat{\text{SE}}$ with $z_{0.975} = 1.959964$.

Source	$\hat{\kappa}_F$	$\widehat{\text{SE}}$	Wald 95% CI
<code>misskappa</code>	0.430245	0.053288	[0.325802, 0.534687]
<code>irrCAC</code>	0.430245	0.054199	[0.324017, 0.536472]

Scaling: the standard errors differ by a factor 1.017095, which equals $\sqrt{n/(n-1)} = 1.017095$. This suggests `irrCAC` uses a $1/(n-1)$ scaling where `misskappa` uses $1/n$. Also, `irrCAC::fleiss.kappa.dist()` prints `conf.int = (0.319, 0.541)`, which corresponds to an implied quantile 2.043499 (close to $t_{n-1, 0.975} = 2.045230$), rather than $z_{0.975}$.

Simulation check (coverage)

We simulate exchangeable raters with $C = 5$ categories and $R = 6$ ratings per subject as follows: draw a latent true label $T_i \sim \text{Categorical}(p)$ and then, for each rater, report T_i with probability $1 - \varepsilon$ and otherwise draw an independent label from p . This implies the population Fleiss kappa equals $\kappa_F = (1 - \varepsilon)^2$ (independent of p). We set $\kappa_F = \hat{\kappa}_F$ from `dat.fleiss1971` so $\varepsilon = 0.344070$, use p equal to the empirical marginal distribution in `dat.fleiss1971`, and evaluate Wald 95% CI coverage over 2000 Monte Carlo replicates (seed 1).

Setting	Mean($\hat{\kappa}_F$)	SD($\hat{\kappa}_F$)	Mean(SE)	Coverage
$n = 30$ (<code>misskappa</code>)	0.417915	0.057713	0.056603	0.9230
$n = 30$ (<code>irrCAC</code>)	0.417915	0.057713	0.057570	0.9285
$n = 2000$ (<code>misskappa</code>)	0.430067	0.007308	0.007116	0.9465
$n = 2000$ (<code>irrCAC</code>)	0.430067	0.007308	0.007118	0.9465

Both implementations agree closely; in particular, their mean standard errors differ by a factor very close to $\sqrt{n/(n-1)}$ (e.g., for $n = 30$, 1.017095 vs 1.017095). For $n = 30$ the (normal-quantile) Wald intervals undercover (around 0.92–0.93 in this experiment), consistent with small-sample bias and/or normal approximation error. For $n = 2000$ the coverage is close to nominal for both.